

Effects of the Integrated Online Advance Organizer Teaching Materials on Students' Science Achievement and Attitude

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Abstract This two-group quasi-experimental study investigated the effects of the Online Advance Organizer Concept Teaching Material (ONACOM) integrated with inquiry teaching and expository teaching methods. Grade 7 students' posttest performances on the light unit achievement and light unit attitude tests controlled for gender, previous semester science grade, and pretest scores were analyzed. No significant treatment effects were found between the inquiry and expository approaches. However, both groups demonstrated significant pretest–posttest gains in achievement and attitude. Independent from the method used, ONACOM was judged effective in both groups as students demonstrated increased achievement and attitude scores. ONACOM has a social and semantic network-aided infrastructure that can be adapted to both methods to increase students' achievement and improve their attitude.

Keywords Concept map · Multimedia/hypermedia · Science education · Social networking · Teaching/learning strategies

Introduction

Classroom instructional technologies have transitioned rapidly over the last 25 years from overhead projectors to LED screens, handheld devices and smart boards. Therefore, inquiry learning in modern educational environments frequently includes crucial technologies: interactive notebooks, Internet-based drawings, spreadsheets, and multimedia or hypermedia systems. It is essential to integrate these technologies and digital strategies in the most effective manner to engage 'net generation' students and to support knowledge construction processes that enhance learners' understanding and attitudes.

The success of infusing technologies and digital strategies into science learning needs to incorporate the lessons learned in less technology-rich environments, the functional capacity of the target technology, and applications across different instructional approaches. Concept mapping, for example, enables teachers to document and use their students' previous schema about the target concepts as foundations for planning and enacting effective teaching (Ausubel 2000). Concept maps are frequently used by teachers as advance organizers to help connect the known and unknown for students and to help them seek, integrate, and organize new knowledge into existing knowledge networks (Jonassen 2006; Novak 2008). Several modern technologies provide rapid visual display of graphic ideas annotated with written text, which is a common experience in the lives of technology-savvy students. Computer-supported concept maps or interactive tools allow students to manipulate virtual contents or objects (Schaal et al. 2010). Teaching materials supported by technology such as the Internet and multimedia increase students' interactivity and motivation and decrease the cognitive load of the learning task. This quasi-experimental study developed and evaluated an Online Advance

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Organizer Concept Teaching Material (ONACOM) that provides online concept maps and facilitates the addition of digital content to the maps and evaluates its influences on middle school students' science achievement and attitude in two different instructional approaches (i.e., inquiry and expository). This pretest–posttest study of achievement and attitudes investigated the following research questions (RQ):

1. Are there any significant differences between the two instructional approaches?
2. Are there any significant gains in the students' achievement and attitude using the ONACOM-integrated inquiry teaching method and the ONACOM-based expository teaching method?

Background

Concepts are used as key elements to construct scientific knowledge as a network of related ideas. However, it is ineffective to learn new knowledge by simply learning isolated concepts; rather, new knowledge should be constructed, adapted, and connected to existing knowledge (Kern and Crippen 2008). A conceptual network can be grown with additional examples related to established concepts or changed by drastically reorganizing and creating new relationships. The conceptual change view of learning involves retrieval of stored ideas from long-term memory, input of external sensory experiences, and information from short-term memory into working memory, where either new knowledge is constructed or existing knowledge is confirmed and then stored in long-term memory.

Concept Mapping and Multimedia Concept Maps

Effective retention and retrieval of knowledge related to long-term memory is mostly the consequence of meaningful learning rather than rote learning (Chiou et al. 2015; Novak and Cañas 2008). Because of the limited converting feature and capacity of short-term memory, if the input information (e.g., visual and audio stimuli) receives the learner's attention, then the integration of new and existing knowledge can be achieved and learning becomes more meaningful. Concept maps are hierarchical and horizontal integrated visual displays of concept nodes and relationships among these nodes within a conceptual network. The features of concepts maps can enhance learning, thereby allowing students to compare their existing and new knowledge (Novak and Cañas 2008). They provide an effective assessment of students' prior knowledge, misconceptions, and acquisition and accommodation of new knowledge (Kern and Crippen 2008).

However, Mayer (2009) expressed concern about the lack of concept maps in science inquiry learning. In cognitive theory of multimedia learning, he believed that students learn better when words and pictures are presented together in a multimedia presentation. Classical dual-coding theory also supports that in multimedia learning environments knowledge is often obtainable in various codes like verbal and visual elements (Paivio 1986). Moreover, students may construct more accurate scientific knowledge when they know the names and characteristics of the concepts (Mayer 2009; Pöhl and Bogner 2012) and the cognitive load can be managed by connecting the learner's prior knowledge to the target concepts (de Jong 2010). These cognitive ideas clarify the basic structural perspective underlying the use of ONACOM.

Well-planned integration of technology with course content and evidence-based classroom practices improve students' conceptual understanding (Çalik et al. 2014; Linn and Eylon 2011). Studies have indicated that concept learning should encourage students' scientific inquiry. This would also include the use of innovative computer/Internet-based technologies (Akpınar 2014; Çalik 2013; Çalik et al. 2014; Ebenezer et al. 2012), dynamic visualizations through web-based inquiry environments (Levy 2013; Zhang and Linn 2011), virtual science laboratories (Kozhevnikov et al. 2013), interactive concept map-oriented mobile learning (Hwang et al. 2011), animations with hypermedia environments (Ebenezer 2001), computer-based and multidimensional concept-mapping tools (Çepni 2009; Chiou et al. 2012; Huang et al. 2012; Yen et al. 2012), and multimedia concept-mapping tools (Asan 2007; Tergan et al. 2008).

The combination of multimedia and concept maps is supported by the studies of CmapTools (Novak and Cañas 2008). This software package was initially developed to allow users to prepare concept maps and to add digital content to the concepts. Information can be visualized clearly, and students have opportunities to think, analyze information, and synthesize knowledge. Later, the software was enriched with multimedia content that could enhance the concepts on the maps. CmapTools has cloud features, but it is not open source, does not provide interactive opportunities for users, and lacks properties such as semantic or social networks since it is a downloadable software package. However, there are other web-based concept-mapping tools like Think Composer (<http://www.thinkcomposer.com>) that allow users to add pictures to concepts and enable mindmapping or drawing flowcharts. Compendium NG (<http://www.compendiumng.org>) is a downloadable application that provides an adaptable visual interface for managing the connections between information and digital resources. Inspiration (<http://www.inspiration.com>) is a web-based concept-mapping tool with the advantage of a graphic organizer feature.

Advanced computer-aided concept maps allow information to be visualized, which decreases the students' cognitive load and supports their knowledge management (Tergan 2005; Tergan et al. 2008). Asan (2007) used Inspiration in a study of Grade 5 students and found that, when information was supported visually and verbally, ideas were recalled much better. He indicated that concept maps reinforced the students' learning, contributed to their achievement, and supported both constructivist learning and teaching.

Recent technology advances (e.g., mobile applications, smartphones, and mobile devices) support the use of web-based materials. Huang et al. (2012) studied three groups of Grade 4 Taiwanese students taught with different materials: a multidimensional concept map, a traditional concept map, and a course book (traditional teaching). Students in the multidimensional concept map group achieved higher outcomes than the other two groups, and students in the traditional concept map group achieved higher outcomes than the group that received a course book approach.

Yen et al. (2012) found that two groups of university students in Taiwan using an Android application with either text or picture concept mapping demonstrated positive effects on their learning outcomes and cognitive processes while gaining computer skills. Although there was no significant difference between the two groups' achievement, the picture-based concept-mapping group had better performance than the text-based concept-mapping group on comprehension and cognitive level of creativity. Chiou et al. (2015) studied two groups of students in which one group used a digital teaching materials approach and the other group used multimedia animation and a multidimensional concept map. The latter group outperformed the former in learning achievement, satisfaction, and retention. They indicated that the latter group constructed their own cognitive structures and efficiently transferred their knowledge into practice. Çepni (2009) examined the effect of using computer-aided teaching tools on Grade 4 students' concept construction and clarification of misconceptions in a light, light sources, and seeing unit. He enriched the concept mapping by putting pictures of related objects under some concepts; he argued that this enriched map supported the students' learning since it helped them construct concepts more clearly and gave them scientific interpretations of the concepts. Technology-infused learning environments might provide the support to improve students' concept maps and their interpretations. Albayrak and Yildirim (2015) found that using social networking sites improved the interaction between the instructor and university students and even among the students. The materials or tools described in the literature lacked semantic and social networking features that, if well

integrated into the online resource, could motivate students to be more cooperative.

Online Advance Organizer Concept Teaching Material (ONACOM)

The conceptual change view of learning, the contributions of coordinated verbal and visual information, and the empirical results of prior studies provide the foundation for the development of an online resource in an attempt to address the lack of fully online features and to serve as a new resource for science teaching. The ONACOM is an original resource that has semantic properties and a specific database structure written in HTML-5 script. Science teachers using ONACOM can enter the related concepts and relational words to combine concept maps in a certain order. They can drag and drop the concept boxes on the map editor screen and then connect the concepts as desired. They can also add a digital content related to the concepts (e.g., online educational videos, animation simulations, pictures, games, activities, experiments, stories, misconceptions, and concept cartoons) that have been loaded into the system. If teacher-users cannot find any relevant content in the existing menu, they may load new content and connect it to as many of the digital nodes as desired and appropriate. Every map has scientific information links that students can use to widen their knowledge about the given subject. The social networking feature of ONACOM enables sharing digital content among teachers and between students and teachers. Moreover, any map that is used by one teacher can be shared with other teachers who are members of the ONACOM community. The content and maps can only be changed by the creator-teacher or other teachers who the creator authorizes. Students, though, do not have permission to change or edit the structure or content links of the map.

Each teacher, student, and administrator has a personal profile page. The ONACOM with Turkish abbreviation 'ÇİDKOM' has an advance organizer program in Turkish (cidkom.mehmetakif.edu.tr or www.cidkom.com) that offers the following innovations for teachers:

- See and share concept maps and related contents that they prepare,
- Easily distinguish the content that has been added,
- Prepare many different concept maps, install content on them, and connect them with each other,
- Share content and maps with all members of an authorized community of users,
- Message other members and post announcements,
- Share maps (in picture format) with members or nonmembers,

- Use the instant messaging service to message students in the class,
- Follow and report what (e.g., time, contents, map) the students have studied, and
- Comment on maps that have been shared.

The focus group for the trial evaluation of ONACOM was middle school students, where an uncontrolled social networking feature was not appropriate. The social networking infrastructure was under the control of the teacher who could adjust the time duration of instant messaging and could confirm or erase irrelevant comments on the maps. Students were able to do the following:

- Like and comment on maps,
- Evaluate the content they watched, read, or followed,
- Surf through content that is controlled by the teacher,
- Use the instant messaging function to message other classmates or the teacher,
- Access the program using any device with an Internet connection, and
- See statistical information on maps.

Figure 1a illustrates the critical conceptual nodes in the topic about lenses within Fig. 1b of the light unit. When young students see the word “lenses” or when their teacher tells them “lenses,” it is only a word to them. It is actually the label for a reasonably complex concept in science (e.g., lenses), but no one can be sure what the students understand when they see the word. Even when students have used lenses before, the teacher should use objects (e.g., a picture, symbol, video, story, or sample lenses in various forms) that anchor the verbal descriptions to make the learning

meaningful. The “lenses” concept is, therefore, supported with a video, simulation, experiment, and photographs. When the cursor is moved to the video icon, the names of the available videos appear in a pop-up window. The digital contents are hyperlinked to the original documents presented in various web pages; when users click on a hyperlink, it can be viewed on a new screen of the ONACOM portal rather than in a new web page in the browser.

In the current study, the basic design procedure for the ONACOM was rapid prototyping—a software development model found to be effective and efficient because of the reduction of project cycle, product quality, and customer and designer satisfaction (Jones and Richey 2000). This procedure has several phases: assessing needs and analyzing content, setting objectives, constructing or designing a prototype, utilizing and researching the prototype, and installing and maintaining a system (Tripp and Bichelmeyer 1990). The development of ONACOM focused on the needs of students to learn and of teachers to teach about the relationships among light, lenses, and seeing, identified the learning objectives, developed the prototype then utilized experts and students to evaluate the prototype, and identified weaknesses and issues related to the prototype in order to revise the prototype.

Methods

Research Design

A two-group pretest–posttest quasi-experimental design was used to analyze the relative effectiveness of the two

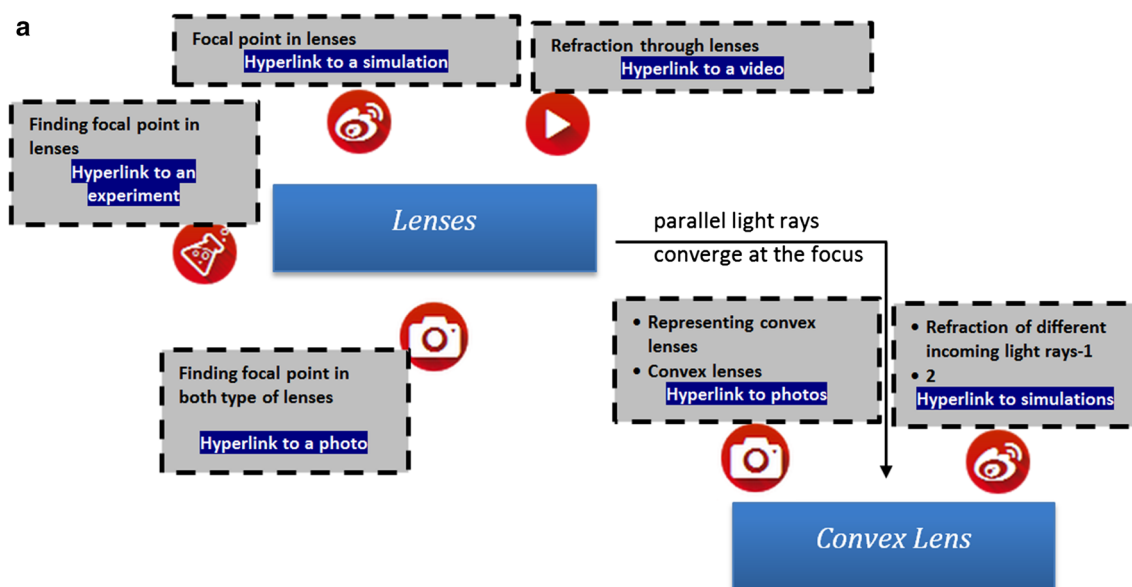


Fig. 1 Draft map of the lenses segment of ONACOM (a) and the original map from the ONACOM including the draft section (b)

b

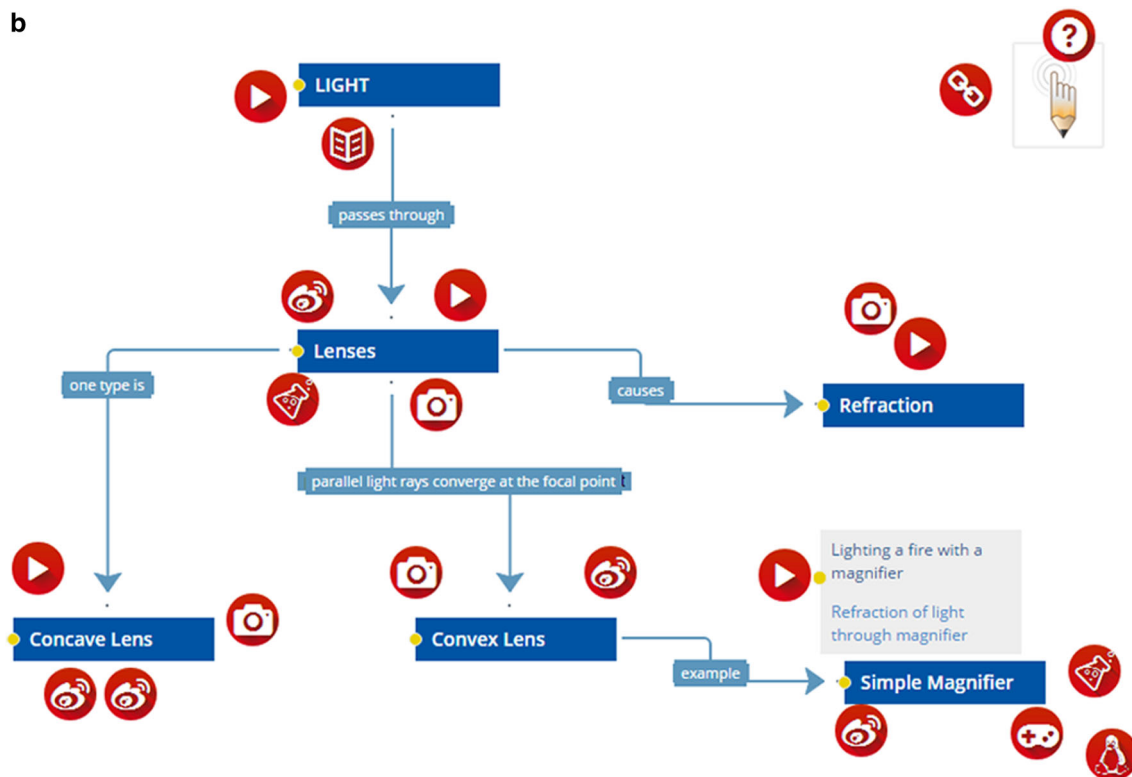


Fig. 1 continued

Table 1 Quasi-experimental research pattern of the study

Group	Pretest	Method	Time period	Posttest
Experimental	LACH, LATT	ONACOM-integrated inquiry teaching method (5E model)	4–5 weeks (16 class hours)	LACH, LATT
Comparison		ONACOM-based expository teaching method		

LACH light achievement, LATT light attitude, 5E engage, explore, explain, elaborate, evaluate

distinct teaching methods with an integrated ONACOM on Grade 7 students' posttest achievement in and attitudes toward a light unit. Students' gender, previous semester science grade (PSG), and pretest achievement and attitudes scores were used as potential covariant and blocking variables. The experimental method is summarized in Table 1.

Population and Sample

This study involved Grade 7 students during the spring semester in the Burdur city-center of Turkey. Seventh grade was selected because it has the greatest number of learning outcomes for a light unit. Light was chosen as the topic because students often have strong misconceptions about it and need to use Internet-based sources more than they do for other concepts. Science teachers also need compact teaching materials that can be easily manipulated

in class and Internet resources for teaching concepts about light using a variety of instructional approaches.

All procedures performed in this study involving students and teachers were in accordance with the ethical standards of the Institutional Research Committee of Mehmet Akif Ersoy University. At the time of the investigation, there were approximately 520 Grade 7 students in 12 schools in the Burdur city-center. Since the research procedure called for applying two teaching methods in two different classes within the same school environment, all of the schools that had two Grade 7 classes were selected as a representative sample. Five schools met this criterion; two schools had more than two Grade 7 classes, and therefore, two classes were selected randomly; in the other three schools, all the Grade 7 classes were selected to participate. One class of students in each school was randomly selected as the inquiry group, leaving the other class as the expository group. The pretest was given to 335 students and the

posttest to 339 students. However, some students' data could not be matched because the tests lacked a student nickname or were completely empty. Therefore, the data of 300 students (158 male and 142 female) from the five schools were analyzed in the study.

The experimental (inquiry) group consisted of 76 male and 72 female students, and the comparison (expository) group consisted of 82 male and 70 female students. Thus, the gender distribution was similar across the two groups. There were no comparisons within or between schools to avoid any school-specific effects or teacher identity. With respect to other demographic variables, these students' average PSGs were 75.73 for the experimental group and 75.08 for the comparison group, both of which are considered "good" (medium: 55–70, good: 70–85, very good: 85–100) within the local standards. A Chi-square analysis indicated that there were no significant differences between the two groups in the categorical variables of gender, $\chi^2(1, N = 300) = .203$, $p = .653$, and PSG, $\chi^2(1, N = 300) = 251.125$, $p = .258$.

Assessment Instruments

The two instruments used as pretests and posttests were a light unit achievement test (LACH), which was prepared for this study, and a light unit attitude scale (LATT), which was adapted for this study. Both tests were tested through a pilot study that included 337 Grade 8 students. The data were analyzed and filtered, so the final analysis of the pilot study for the LACH included the data of 318 students and for the LATT 326 students.

The Achievement Test

The researchers compiled the LACH from the related literature, paying attention to the questions covering all learning outcomes for the Grade 7 level, including the light unit. The first draft of the LACH contained 38 questions selected from a pool of 120 questions; it was prepared by the researchers who considered the unit's content and analyzed questions from the Level Placement Tests (SBS) of previous years as well as various textbooks. Necessary changes were made to the questions and answer choices to cover and match the learning outcomes. Face validity was explored by five academicians, seven science teachers, and a Turkish language teacher. They analyzed the initial draft to evaluate the appropriateness of the questions' form and content as well as proper grammar usage. The construct validity was explored through a pilot study by giving the test to Grade 8 students who had learned about light concepts previously. The ITEMAN (<http://iteman.software.informer.com/4.1/>) was used to determine reliability of

item deleted values, item discrimination index, and point-biserial values for the test results from the pilot study.

The LACH test items answers were scored as correct (1) or incorrect (0). The total achievement score for each student was transformed into a percentile score. The discrimination power and difficulty index were analyzed for each item; those whose values were below .200 (items 9, 15, 22, 23, 25, 29, and 35) were revised and improved. Expert views were also provided for the final form of the LACH. The mean of the pilot study was 54.0, the standard deviation was 4.78, and the maximum and minimum scores were 82.8 and 22.0. The Cronbach's alpha reliability coefficient was .71 in the pilot study. The final form of the LACH included three sections: true/false (10), matching (5), and multiple-choice (23).

The Attitude Test

The attitude test was designed around the Simple Electric Circuit unit (Taslidere and Eryilmaz 2012). In this study, it was adapted by substituting the subject light for electricity. The LATT includes 24 positive 5-point Likert-type items (5 = strongly agree ... 1 = strongly disagree). The total attitude score was obtained by adding the numerical scores of all items for each student. The same experts who evaluated the LACH items checked the items in the LATT to ensure face validity.

Construct validity of the LATT was explored by administering it to 326 students in the pilot study of the LACH. An explanatory factor analysis was carried out by principal component analysis as the extraction method and varimax with Kaiser normalization as the rotation method. Verification of data adequacy was done with the Kaiser–Meyer–Olkin (KMO) and the Bartlett's sphericity tests. The KMO value was found to be .936, which is accepted as "very good" in order to extract factors from the collected data (Büyüköztürk 2010). The Bartlett's sphericity test result was $\chi^2(276) = 4957.659$, $p < .01$. Therefore, the data obtained from the LATT were suitable for factor analysis.

Five factors with eigenvalues of 1.0 or higher were extracted by SPSS. Together, they accounted for 69 % of the variance, which is accepted as high (Büyüköztürk 2010). Furthermore, the items in the five factors had loadings between .306 and .810, which is above the critical limit of .300. The factors were analyzed in terms of loaded items (Table 2), and they were later named using the category names from the literature (Taslidere and Eryilmaz 2012). Revised LATT items 9, 15, 19, and 24 loaded on different factors than reported by Taslidere and Eryilmaz (2012); however, the internal consistency of the five factors (Cronbach's $\alpha = .867, .902, .885, .873$, and .835,

Table 2 Items in light unit attitude scale and their categories

Factor	Item number	Attitude category
1st	9, 11, 12, 13, 14, 24	Interest-related behavior
2nd	1, 2, 3, 4, 5	Interest (like)
3rd	19, 20, 21, 22, 23	Self-efficacy
4th	15, 16, 17, 18	Achievement motivation (their effort to achieve)
5th	6, 7, 8, 10	Importance (future expectations)

respectively) and the whole test ($\alpha = .945$) were considered to be acceptable.

Comparison and Experimental Group Applications

Clark (1983) indicated that the general approach to the effectiveness of media studies should be toward integrating the media into both experimental and comparison group applications. In the comparison group, the ONACOM is a suitable material that can be used as an advance organizer at the beginning of each lesson. Ausubel (2000) suggested that in effective expository teaching it is crucial to establish a meaningful connection between new and existing information in expository learning and teaching settings. The use of advance organizers is a crucial part of this method.

In the experimental group, students' inquiry was supported with the ONACOM by incorporating model progression, assignments, and a shared proposition tool into the simulation within the 5E model. The 5E model (engage, explore, explain, elaborate, evaluate) is a constructivist approach believed to reflect a learner-centered environment; its effectiveness for student learning is based on their active participation in the teaching/learning process (Gijlers and de Jong 2013; Kanlı 2007).

Before the applications were enacted, both the experimental and comparison group teachers were trained for approximately one month in terms of the usage of ONACOM and its integration into their teaching methods. The students were also informed about the method that they would use. The lessons were conducted in their normal classroom setting. First, the pretests were administered to the students; at the end of the week of applications, they completed posttests. After the teachers had started their class in each group, they and an assigned researcher delivered the activity sheets to the students. The applications that covered the light unit, including learning outcomes, were related to the subjects of absorption of light, seeing colored objects, transmission of light from one medium to another, and lenses. All ONACOM concept maps with appropriate digital content were prepared before application in both groups. In the experimental group, the implementations were preferably done in computer laboratories where there was one computer per student. The comparison group applications mostly took place in the

normal classroom environment with projectors and teacher computers. The applications that were completed in 4–5 weeks (more than 16 class hours) in both the comparison and experimental groups are briefly described below for two class hours ($40 + 40 = 80$ min).

Comparison Group Applications (Expository Teaching)

The class session was divided into four stages: introduction, advance organizers, lectures, and strengthen the cognitive organization. The map of the lesson was shared with the students at the beginning of the lesson.

1. The *introduction* was planned to take 5 min. The teachers explained the general goal of the class and what the students would learn. They then asked questions from the activity sheet to gain the students' attention and clarify the expected answers. Since the questions were related to real-life topics, they gained the students' attention and many students raised their hands to answer.
2. The presentation of *advance organizers* was planned to take 10 min. In this stage, the teacher displayed the concept map of the corresponding topic in ONACOM, the concepts attributed to the topic and all digital (online/offline) content uploaded to these concepts, including scientific information (chain symbol connected to the ONACOM logo) and evaluation questions (question mark connected to the ONACOM logo). For the concept of "magnifier," there were multiple icons for each of which more than one digital content could be uploaded. For example, under the video icon there were two videos "refraction of light through magnifier" and "lighting a fire with a magnifier." By moving on an icon, the name of the contents appeared in a box and when the name was clicked, a hyperlink opened in another window of ONACOM. All of the teachers' instructions and explanations were from the ONACOM content. For example, the teachers asked students to examine the concept map and relations of the concepts and to write three sentences (e.g., "*light is refracted* when it passes through *lenses*") on the activity sheet reflecting the concepts and relations.

3. The *lecture* stage took almost 50 min because the teachers were presenting a new topic. After the students had roughly analyzed the map, the teachers asked them to listen carefully while they opened a scientific information file (chain symbol connected to the ONACOM logo). All examples, presentation files, videos, and so on demonstrated by the teachers were taken from the digital content loaded to the ONACOM map. The teachers ensured the students saw different examples and then asked them to review what they had learned in class and solve the problems on the back of the activity sheet. The same problems were presented in ONACOM, projected on a screen, and solved in the class.
4. In the *strengthen the cognitive organization* stage, which took 15 min, the teachers opened the map and asked the students to review the relations between the concepts to strengthen their understanding. After that, the students examined any digital content that could not be analyzed before or was crucial to their understanding. Finally, the teachers asked the students to complete an empty concept map.

Experimental Group Applications (5E Model)

In the experimental group, the learning cycle (5E model) was used, which had been verified during implementation. Therefore, in these groups, the classes covered the five stages.

1. *Engage* (15 min): First, the teachers asked the students questions from the activity sheet they had been given. Since the goal was to arouse curiosity, the same question as in the comparison group was used. Then the teachers opened the first part of the map and shared it with the students. The whole map was the same as the map used in the comparison group, but it was not given to the students in its entirety since the 5E model requires students to construct their own knowledge. Therefore, the map was divided into three parts. At first, the students were asked to examine the digital content (video, story, pictures, etc.) loaded to the main concept *light* for 3–5 min. The teachers asked the students to fill in the blanks on the activity sheet.
2. *Explore* (25 min): No scientific information was provided directly to students during the exploration. By examining the map, they explored details of the lenses and answered the corresponding questions on the experimental group activity sheet. Moreover, teachers asked students with low motivation directive questions like “What do you think about the answer to this question? If you want, you can look at more digital content on ONACOM.” Students were provided sufficient time to find answers to the questions.
3. *Explain* (20 min): The majority of students in the class were asked to provide generalization sentences and then to share their answers to the questions. Students who were not able to find answers had an opportunity to learn more information by working with their peers. The teachers asked them to finalize their incomplete answers. At this point, the second part of the map was opened and shared on ONACOM, and the students were asked to examine the contents. In this stage, they were also asked to find answers to any questions they had thought of themselves. Following this, the teachers shared the entire ONACOM map and asked the class to examine the scientific information file. They asked students to write down the similarities and differences between the scientific definition or information and the knowledge that they had constructed on the corresponding spaces of the activity sheet.
4. *Elaborate* (10 min): The students continued to use ONACOM to find answers to new questions given in the activity sheet and were asked to find answers to new questions about lenses directly related to a new real-life situation, either individually or in collaboration with their peers. They were encouraged to offer as many solutions as possible to the new questions by using the related contents of ONACOM.
5. *Evaluate* (10 min): The teachers presented the questions that were uploaded to ONACOM and projected on a screen as well as available on the back of the activity sheet. Moreover, the students were free to use all the digital content and information and to reach a consensus through peer discussions. Every student was provided an opportunity—sometimes students were chosen by the teacher—to talk about their findings.

Data Collection

The LACH and LATT were administered to the students in the middle and at the end of the spring semester. Students were asked to write a nickname on their test sheets to match their pretest and posttest scores. The tests took approximately 45 min in total. The nature of this research pattern included various possible threats to internal validity (e.g., location, data collector characteristics/bias, attitude of participants, maturation, and implementation). The same teacher taught the content to both groups in natural classroom settings of each school within the same teaching period; as well, the students took the same instruments in order to compare the potential influences of the threats.

Data Analysis

The data analysis considered the group (experimental vs. comparison) to be the independent variable. The data for the experimental and comparison groups across all schools were combined into a single dataset since the focus of this study was the two treatments, not a comparison among schools or teachers. The pretest and posttest data were analyzed to provide descriptive statistics, explore the similarity between groups prior to study, and explore gains and differences between instructional approaches upon completion of the study. The students' gender, PSG, pretest achievement and pretest attitude results were considered as independent variables or covariates. Descriptive statistics were calculated using standard procedures. Achievement scores were on a basis of 100 and ranged from 0 to 100. Attitude scores were collected on a 5-point Likert-type scale. Attitude scores ranging from 24 to 44 were coded as very negative, from 44 to 63 as negative, from 63 to 82 as neutral, from 82 to 101 as positive, and from 101 to 120 as very positive.

The PSG and pretest achievement and attitude data were explored for similarity between the combined groups at the outset of the study using *t* tests. Based on the *t* test results, it was decided to use a multivariate analysis of covariance (MANCOVA) to analyze the posttest data since there were differences between the PSG and pretest data for the combined experimental and comparison groups.

Descriptive and inferential statistics were calculated using SPSS 22. The assumptions of *t* test and MANCOVA were analyzed. The MANCOVA was applied separately to the posttest achievement and attitude as dependent variables and gender; the PSG and pretest achievement and attitude were used as covariates with respect to the two groups where appropriate. Error types, power, and effect sizes (*d* and η^2) were evaluated using established principles (Field 2005; Fraenkel et al. 2012). Therefore, the estimated required sample size was calculated as $N = 288$ following Cohen's (1988) approach. The sample ($N \sim 300$) for this study meets the estimated sample size.

Results

The main research focus of this study was: "Are there any significant effects of the ONACOM-integrated inquiry teaching method in comparison with the ONACOM-based expository teaching method?" The question was explored by analyzing students' posttest achievement scores from the LACH (POSTLACH) and posttest attitude scores from the LATT (POSTLATT) in a Grade 7 science course when their gender, PSG, pretest achievement scores from the LACH (PRELACH), and pretest attitude scores from the

LATT (PRELATT) were controlled. The descriptive statistics were calculated, and a preliminary analysis for inferential statistics was conducted to justify the final analysis. The MANCOVA was used to evaluate any differences between the inquiry and expository approaches. This test was performed because it can equate groups of one or more independent variables—in this case, the students' gender, PSG, PRELACH, and PRELATT—and can be used for more than one dependent variable—in this case, the POSTLACH and POSTLATT. The multivariate outliers were identified by examining Mahalanobis values. Mahalanobis distance values are evaluated based on Chi-square critical values of the study sample, and the cases (for participants 63, 243, 244, and 280) having significant values ($p < .05$) excluded from further analysis (Byrne 2013; Field 2005). Table 3 provides summaries of the Kolmogorov–Smirnov test, skewness, and kurtosis values for both the attitude and achievement scores.

Inspection of the statistics in Table 3 indicates that the average POSTLACH of the experimental group was greater than that of the comparison group, while the average POSTLATT was greater in the comparison group than in the experimental group. In general, both groups (independent of the method) had medium level achievement (between 55 and 60) and positive attitudes (between 82 and 101). Since the sample size is large, the normal distribution with Kolmogorov–Smirnov analysis may fail; however, the skewness and kurtosis values (between -1 and $+1$ for all) indicate there is no violation of normal distribution. Finally, to control whether the further assumptions of MANCOVA were satisfied, preliminary assumption testing for homogeneity of regression, linearity, sphericity, equality of variance, multicollinearity, and independency of observations were also conducted; no significant violations were found.

RQ1: Are there any significant differences between the two instructional approaches?

The POSTLACH and POSTLATT were attained as dependent variables. The independent variable *group* was attained as a fixed factor. The variables of gender, PSG, PRELACH, and PRELATT were checked in terms of whether they satisfied the conditions of covariate. Raykov and Marcoulides (2012) accepted gender as a time-independent covariate even though it is categorical. There was no significant interaction between any of the covariates and the independent variable. Another criterion for covariates is that they should correlate with one of the dependent variables. The values of the Pearson correlation coefficients revealed that the four variables correlated with at least one of the dependent variables, indicating that they satisfied the condition of being covariate. The equality of covariance matrices was also controlled by Box's M test. The test

Table 3 Descriptive statistics for the variables

Variable	Group	Kolmogorov–Smirnov			Skewness	Kurtosis	<i>M</i>	SD	Mdn
		Statistics	SD	<i>p</i>					
PRELACH	Experimental	.051	148	.200	−.021	−.576	42.16	8.77	42.20
	Comparison	.080	152	.020	−.177	−.291	39.83	9.08	41.20
POSTLACH	Experimental	.070	148	.076	−.164	−.722	59.11	16.33	59.30
	Comparison	.084	152	.011	−.393	−.295	57.45	14.78	59.40
PRELATT	Experimental	.071	147	.064	−.475	−.115	86.99	17.58	90.00
	Comparison	.500	149	.200	−.743	1.500	89.03	16.19	89.00
POSTLATT	Experimental	.059	147	.200	−.368	−.029	91.93	15.70	92.00
	Comparison	.073	149	.051	−.562	.122	92.60	15.86	94.00

N = 296. PRELATT and POSTLATT scores were 24 (min.) and 120 (max.); PRELACH and POSTLACH scores were 0 (min.) and 100 (max.)

suggested that the covariance matrices were equal and Box's $M = 1.442$, $F(3, 15,649,253.49) = .447$, $p = .698$. Thereby, the MANCOVA was conducted, and those results are reported in Table 4.

Inspection of the results in Table 4 revealed that there was no statistically significant group effect on the POSTLATT or POSTLACH. In other words, there was no significant difference between the inquiry-based experimental and expository-based comparison groups. The result indicates the need for further exploration of the learning gains to determine the effectiveness of ONACOM independent of the overall instructional method in which it was embedded.

RQ #2: Are there any significant gains in the students' achievement and attitude using the ONACOM-integrated inquiry teaching method and the ONACOM-based expository teaching method?

The second RQ was analyzed by a series of paired sample *t* tests. Analysis of the experimental group's pretest and posttest achievement data indicated that there were significant mean differences between the PRELACH ($M = 42.16$, $SD = 8.77$) and POSTLACH ($M = 59.11$, $SD = 16.33$; $t(146) = -13.307$, $p < .001$). The effect size was large (Cohen's $d = 2.19$), and the statistical power was high (.99). Therefore, the effect of the ONACOM-integrated inquiry learning method on achievement was

large; and there was a 16.94-point mean difference between the pretest and posttest scores. Analysis of the experimental group's pretest and posttest attitude data revealed significant mean differences between the PRELATT ($M = 86.98$, $SD = 17.58$) and POSTLATT ($M = 91.93$, $SD = 15.70$; $t(146) = -3.450$, $p < .001$). The effect size was medium (Cohen's $d = .57$), and the statistical power was high (.93). Therefore, the 4.95-point mean score difference between the PRELATT and POSTLATT scores was significant in the experimental group with ONACOM-integrated inquiry learning.

Analysis of the comparison group's achievement data indicated a significant mean difference between the PRELACH ($M = 39.83$, $SD = 9.08$) and POSTLACH ($M = 57.45$, $SD = 14.78$; $t(148) = 14.448$, $p < .001$). The effect size was large (Cohen's $d = 2.7$), and the statistical power was high (.99). Therefore, the 17.73-point mean difference between the PRELACH and POSTLACH was significant in the comparison group since the effect of the ONACOM-based expository teaching method on achievement was large. Analysis of the comparison group's attitude data indicated a significant mean difference between the students' PRELATT ($M = 89.03$, $SD = 16.19$) and POSTLATT ($M = 92.60$, $SD = 15.86$; $t(148) = -2.422$, $p < .05$). The mean difference was 3.57 points, the effect size was medium (Cohen's $d = .40$), and the statistical power was not high (.67).

Table 4 Results of MANCOVA

Source	Wilks's Λ	<i>F</i>	<i>p</i>	Partial η^2	Observed power
Intercept	.781	40.529*	<.001	.219	.999
Gender	.994	.815	.443	.006	.189
PSG	.609	92.964*	<.001	.391	.999
PRELATT	.838	27.994*	<.001	.162	.999
PRELACH	.984	2.396	.093	.016	.482
Group	.999	.184	.832	.001	.078

N = 296. Hypothesis SD = 2, Error SD = 289

* $p < .05$

Discussion

The ONACOM-integrated inquiry teaching method increased students' achievement and improved their attitude, as did the ONACOM-based expository teaching method for students in the comparison group. The attitude score mean difference of the experimental group was slightly higher than that of the comparison group. This is most likely because the students in the experimental group might have been more enthusiastic in learning the concepts with the integration of ONACOM in the 5E approach. Therefore, their attitude might have been more positive because they liked constructing scientific knowledge and were more active in doing so. Clark (1983) indicated the general approach to the effectiveness of media studies should be toward integrating media into both experimental and comparison group applications. The aim of this study was to understand whether the change was because of the method applied or the media (ONACOM). Clark recommended that, in order to conclude that there is an increase in achievement in a study, the media should be carefully adapted to the instructional methods to prevent uncontrolled effects of the instructional method. The results of studies by Gijlers and de Jong (2013) with inquiry teaching, by Hwang et al. (2011) with a concept map-oriented mobile tool, and by Huang et al. (2012) using a multidimensional concept map demonstrated that students who used these tools had higher achievement than their counterparts in comparison groups who were taught through traditional tools or methods. However, the current study is somewhat different than these studies in that it did not reveal any significant treatment effect. The previous studies implemented the selected teaching material only for the experimental group instead of equalizing all conditions except the instructional method for both groups. In the present study, instead of a traditional method, a well-defined expository teaching method was used in the comparison group. It is expected to be effective when used extensively within any classroom condition by implementing the steps and procedures considerate of grade level and the science unit because ONACOM was used in both groups and it fits the fundamental structures and functions of both the inquiry-oriented and expository-oriented methods.

The students could ask their teacher questions related to concepts about light from their own profile pages using the instant online messaging or the offline messaging feature of ONACOM. The concept-mapping tools discussed in the related literature lacked this feature. The present study indicates that the social networking infrastructure of ONACOM can increase teacher–student interactions, which may encourage students to develop more positive attitudes toward science. Therefore, ONACOM likely increased students' achievement since it attracted their

attention and they used it frequently since it used contemporary logic of social networking, which they enjoy using in other contexts. Albayrak and Yildirim (2015) implied similar results for university students' in- and out-of-class participation by using a well-known social networking site. In the current research, the activities of the users (i.e., students and teachers) were followed during the applications; and it was found that the users continued to use ONACOM at home. The students liked the concept maps prepared for the light unit, shared digital content with their teachers, and wrote messages; and the teachers answered the students' questions within the ONACOM instant messaging system. This feature of ONACOM makes concept learning in science interactive, attractive, and easy to follow.

This study revealed the usefulness of ONACOM. According to Kılınç (2007) and Ausubel (2000), the advance organizer feature of ONACOM helps students construct knowledge faster. Similar graphic organizer programs illustrate similar effects on students' knowledge construction (Asan 2007; Chiou et al. 2015). The finding related to the increase in achievement is supported by the results of Çepni (2009), who indicated that students learn easily with computer-based concept-teaching materials since they can build knowledge meaningfully. It is also supported by the results of previous studies that indicated allowing concepts to be visualized in content-loaded concept maps, digital teaching materials, or multidimensional concept maps results in an increase in the functionality of the maps and makes knowledge acquisition easier for students (Chiou et al. 2015; Huang et al. 2012; Hwang et al. 2011; Pöhl and Bogner 2012; Tergan 2005; Yen et al. 2012).

Conclusion

ONACOM is a concept-teaching material based on a specific science subject in this study, and it can be adapted to any science subject. This study adapted it to teaching a light unit to Grade 7 students by integrating it into two methods used in the experimental 5E approach and the traditional expository approach, respectively, to identify its effectiveness. There was a large difference between the pretest and posttest scores on achievement (approximately 17.5-points) and attitude (approximately 4.5-points), and they did not differ with respect to the comparison versus experimental group. Overall, it might be stated that ONACOM, when implemented well in science teaching by following the procedures described in this study, was effective in the two methods applied to the groups in terms of increasing students' achievement and improving their

attitude. The investigation supported the effectiveness of ONACOM with improvements in the students' attitude and achievement. The use of the ONACOM increased achievement most likely because it helped students to construct scientific knowledge with the necessary amount of conceptual media/content. The ONACOM is crucial in terms of several features for students:

- The concepts and digital contents were suitable to the dual-coding process and increased the time of processing of information in the short-term memory.
- The concepts and the variety of loaded digital contents were to be compatible with their cognitive load in order to foster meaningful learning.
- The students could manage their own learning process by interacting with their learning environment, which positively affected their interest and attitudes.
- Students' efforts during their use of ONACOM can be controlled by the teacher.

Teachers could monitor which students opened the shared map and which digital content was clicked on by which student. Teachers selected the digital contents loaded to the concepts, and they prepared concept maps and concept relations. It can be clearly stated that students liked the use of ONACOM as they used its features frequently, shared digital content (e.g., any photos they took) with their teachers, and shared comments or feelings with the other users about the concept maps. This research will continue with an investigation of science teaching with ONACOM for Grade 6 students. Future studies may concentrate on the effectiveness of ONACOM for different grade levels or disciplines.

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Compliance with Ethical Standards

Ethical Approval All procedures performed in this study involving students and teachers were in accordance with the ethical standards of the institutional research committee of Mehmet Akif Ersoy University with the date February 22, 2013, and document #352.

References

- Akpınar E (2014) The use of interactive computer animations based on POE as a presentation tool in primary science teaching. *J Sci Educ Technol* 23:527–537. doi:10.1007/s10956-013-9482-4
- Albayrak D, Yildirim Z (2015) Using social networking sites for teaching and learning: students' involvement in and acceptance of Facebook® as a course management system. *J Educ Comput Res* 52(2):155–179. doi:10.1177/0735633115571299
- Asan A (2007) Concept mapping in science class: a case study of fifth grade students. *Educ Technol Soc* 10(1):186–195
- Ausubel DP (2000) The acquisition and retention of knowledge: a cognitive view. Kluwer Academic, Dordrecht
- Büyüköztürk Ş (2010) Sosyal bilimler için veri analizi el kitabı. [Data analysis handbook for social studies], 11th edn. PegemA, Ankara
- Byrne BM (2013) Structural equation modeling with AMOS: basic concepts, applications, and programming, 2nd edn. Routledge, New York
- Çalik M (2013) Effect of technology-embedded scientific inquiry on senior science student teachers' self-efficacy. *Eurasia J Math Sci Technol Educ* 9(3):223–232
- Çalik M, Ebenezzer J, Özsevgiç T, Küçük Z, Artun H (2014) Improving science student teachers' self-perceptions of fluency with innovative technologies and scientific inquiry abilities. *J Sci Educ Technol* 24(4):448–460. doi:10.1007/s10956-014-9529-1
- Çepni S (2009) Effects of computer supported instructional material (CSIM) in removing students' misconceptions about concepts: light, light source and seeing. *Energy Educ Sci Technol B Soc Educ Stud* 1(1&2):51–85
- Chiou CC, Lee LT, Liu YQ (2012) Effect of Novak colorful concept map with digital teaching materials on student academic achievement. *Proc Soc Behav Sci* 64:192–201. doi:10.1016/j.sbspro.2012.11.023
- Chiou CC, Tien LC, Lee LT (2015) Effects on learning of multimedia animation combined with multidimensional concept maps. *Comput Educ* 80:211–223
- Clark RE (1983) Reconsidering research on learning from media. *Rev Educ Res* 53(4):445–459
- Cohen J (1988) Statistical power analysis for the behavioral sciences, 2nd edn. Lawrence Erlbaum, Hillsdale
- de Jong T (2010) Cognitive load theory, educational research, and instructional design: some food for thought. *Instr Sci* 38:105–134. doi:10.1007/s11251-009-9110-0
- Ebenezzer JV (2001) A hypermedia environment to explore and negotiate students' conceptions: animation of the solution process of table salt. *J Sci Educ Technol* 10(1):73–91
- Ebenezzer JV, Columbus R, Kaya ON, Zhang L, Ebenezzer DL (2012) One science teacher's professional development experience: a case study exploring changes in students' perceptions of their fluency with innovative technologies. *J Sci Educ Technol* 21:22–37. doi:10.1007/s10956-010-9277-9
- Field A (2005) Discovering statistics using SPSS, 2nd edn. SAGE, London
- Fraenkel JR, Wallen NE, Hyun HH (2012) How to design and evaluate research in education, 8th edn. Mc Graw-Hill, New York
- Gijlers H, de Jong T (2013) Using concept maps to facilitate collaborative simulation-based inquiry learning. *J Learn Sci* 22(3):340–374. doi:10.1080/10508406.2012.748664
- Huang HS, Chiou CC, Chiang HK, Lai SH, Huang CY, Chou YY (2012) Effects of multidimensional concept maps on fourth graders' learning in web-based computer course. *Comput Educ* 58(3):863–873
- Hwang GJ, Wu PH, Ke HR (2011) An interactive concept map approach to supporting mobile learning activities for natural science courses. *Comput Educ* 57(4):2272–2280
- Jonassen DH (2006) Modelling with technology: Mindtools for conceptual change, 3rd edn. Pearson, Boston
- Jones TS, Richey RC (2000) Rapid prototyping methodology in action: a developmental study. *Educ Technol Res Dev* 48(2):63–80
- Kanlı U (2007) 7E Modeli Merkezli Laboratuvar Yaklaşımı İle Doğrulama Laboratuvar Yaklaşımlarının Öğrencilerin Bilimsel

- Süreç Becerilerinin Gelişimine Ve Kavramsal Başarılarına Etkisi [The effects of 7E centered laboratory approach and confirmatory laboratory approach on students' improvement of scientific process skills and conceptual achievements] (Unpublished doctoral dissertation). Gazi University, Ankara
- Kern C, Crippen KJ (2008) Mapping for conceptual change: concept mapping activities encourage students to develop scientific understanding. *Sci Teacher* 75(6):32–38
- Kılınç A (2007) Bir öğretim stratejisi olarak kavram haritalarının kullanımı [Use of concept maps as a teaching strategy]. *Yüzüncü Yıl Üniversitesi, Eğitim Fakültesi Dergisi* 4(2):21–48
- Kozhevnikov M, Gurliitt J, Kozhevnikov M (2013) Learning relative motion concepts in immersive and non-immersive virtual environments. *J Sci Educ Technol* 22:952–962. doi:[10.1007/s10956-013-9441-0](https://doi.org/10.1007/s10956-013-9441-0)
- Levy D (2013) How dynamic visualization technology can support molecular reasoning. *J Sci Educ Technol* 22:702–717. doi:[10.1007/s10956-012-9424-6](https://doi.org/10.1007/s10956-012-9424-6)
- Linn MC, Eylon BS (2011) Science learning and instruction: taking advantage of technology to promote knowledge integration. Routledge, New York
- Mayer RE (2009) Multimedia learning, 2nd edn. Cambridge University Press, New York
- Novak JD (2008) Concept maps: what the heck is this? <https://www.msu.edu/~luckie/ctools/>
- Novak JD, Cañas AJ (2008) The theory underlying concept maps and how to construct and use them (Technical Report IHMC CmapTools 2006-01, Rev. 2008-01). Florida Institute for Human and Machine Cognition, Pensacola. <http://cmap.ihmc.us/docs/theory-of-concept-maps.php>
- Paivio A (1986) Mental representation: a dual coding approach. Oxford University Press, London
- Pöhl S, Bogner FX (2012) Learning with computer-based multimedia: gender effects on efficiency. *J Educ Comput Res* 12(4):387–407
- Raykov T, Marcoulides GA (2012) An introduction to applied multivariate analysis. Routledge, New York
- Schaal S, Bogner FX, Girwidz R (2010) Concept mapping assessment of media assisted learning in interdisciplinary science education. *Res Sci Educ* 40:339–352. doi:[10.1007/s11165-009-9123-3](https://doi.org/10.1007/s11165-009-9123-3)
- Taşlıdere E, Eryılmaz A (2012) Basit elektrik devreleri konusuna yönelik tutum ölçeği geliştirilmesi ve öğrencilerin tutumlarının değerlendirilmesi [Developing an attitude scale toward simple electricity circuits subject and evaluation of students' attitudes]. *Türk Fen Eğitimi Dergisi* 9(1):31–46
- Tergan SO (2005) Digital concept maps for managing knowledge and information. In: Tergan SO, Keller T (eds) Knowledge and information visualization. Springer, New York, pp 185–204
- Tergan SO, Engelmann T, Hesse FW (2008) Digital concept maps as powerful interfaces for enhancing information search: an experimental study on the effects of semantic cueing. In: Cañas AJ, Reiska P, Åhlberg M, Novak JD (eds) Proceedings of the third international conference on concept mapping-2008. Tallinn, Estonia, & Helsinki, Finland
- Tripp SD, Bichelmeyer B (1990) Rapid prototyping: an alternative instructional design strategy. *Educ Technol Res Dev* 38(1):31–44
- Yen JC, Lee CY, Chen IJ (2012) The effects of image-based concept mapping on the learning outcomes and cognitive processes of mobile learners. *Br J Educ Technol* 43(2):307–320. doi:[10.1111/j.1467-8535.2011.01189.x](https://doi.org/10.1111/j.1467-8535.2011.01189.x)
- Zhang ZH, Linn MC (2011) Can generating representations enhance learning with dynamic visualizations? *J Res Sci Teach* 48(10):1177–1198